Department of Computing and Mathematics

HPC

Project Title:
Performance Evaluation and Comparison of Serial, MPI, and OpenMP Implementations of Particle Simulation

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"I, Blessen Koshy Sujan, declare that the work titled “Performance Evaluation and Comparison of Serial, MPI, and OpenMP Implementations of Particle Simulation” is my own, except where acknowledged, and has not been submitted elsewhere."

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DATE: 16/05/2024

REPORT

"Supercomputing is the cornerstone of scientific discovery. It transforms the theoretical into the practical, turning the unknown into the known." – Jack Dongarra

In this report, the performance of three versions of particle simulations namely Serial, MPI and OpenMP are evaluated. They are named as 'md.c','md.MPI' and md.OpenMP.c' respectively. The movement of 20000 particles over 10 times are simulated by the program. Also, to determine the correct numerical values and to analyze their performance, parallel scalability and efficiency, the program is compiled and ran each version of it with appropriate optimization.

In the methodology, the initial step is the creation of new directory in the Linux terminal named 'HPCAssessment' using mkdir command. Then the directory is changed to HPCAssessment with the aid of command cd. Later, the file containing the program is extracted to working directory using tar -xf command, which is followed by the viewing the contents of the files in the directory with ls command. This has showed us the contents in the file as md.c , md-MPI.c and md- OpenMP.c.

Serial Implementation is done in the first place. The compilation command for the serial implementation is **'gcc-O0 md.c-lm-o md-serial-O0.exe'**. After that, it is executed with the **'./md'.** This executed the program with the optimization level 0 and we got the time for the simulation of 20000 particles in 10 steps as 126.225 seconds. And the position of the centre of mass of the molecules is found at **(-0.095,-0.166,49.646)** .It is followed by the execution with 1st, 2nd, 3rd and even 4th optimization levels. The values and position of the centre of mass at each level is tabulated below:

SERIAL		
Optimization	TimeStamp	Centre of Mass(x,y,z)
0	126.225	(-0.095,-0.166,49.646)
1	71.1626	(-0.095,-0.166,49.646)
2	58.6083	(-0.095,-0.166,49.646)
3	61.957	(-0.095,-0.166,49.646)
4	60.4489	(-0.095,-0.166,49.646)

- It is found from the table that the performance had increased significantly with in the optimization levels and the best time recorded was at optimization level 2 with 58.6083 seconds for the simulation of 20,000 molecules.
- It can also be noted from the table that, even though, the optimization levels are changed, the centre of mass remained at the same position due to the conservation of momentum.
- Also, a V shaped graph can be produced if the Time stamps are plotted against the optimization levels.

Since the best performing optimization level was 2, it is set as default level during the execution of both the parallel programming MPI and OpenMP.

MPI Implementation is done on optimization level 2, with changing the number of processors. This process is achieved by implementing at optimization level 2 with 'mpicc-O2 md-MPI.c-lm-o MPI.exe. After that, the parallel programming is executed with changing the number of processors from 1 to 12. The simulation is executed at each number of processors and the average time stamp is calculated. The command used for the execution is mpirun-np./MPI.exe. For each number of processors, the execution is done thrice, and the average time is calculated. The average time stamps are shown below.

MPI					
Processor	TimeStamp 1	TimeStamp 2	TimeStamp 3	AVG	Centre of Mass(x,y,z)
1	59.6561	60.4916	56.4395	58.8624	(-0.095,-0.166,49.646)
2	30.2753	28.2463	30.2335	29.58503	(-0.095,-0.166,49.646)
3	0	0	0	0	(-0.095,-0.166,49.646)
4	16.7685	15.1196	14.1795	15.35587	(-0.095,-0.166,49.646)
5	11.4526	11.4111	11.4765	11.44673	(-0.095,-0.166,49.646)
6	0	0	0	0	(-0.095,-0.166,49.646)
7	0	0	0	0	(-0.095,-0.166,49.646)
8	11.1176	11.0095	11.1411	11.0894	(-0.095,-0.166,49.646)
9	0	0	0	0	(-0.095,-0.166,49.646)
10	9.027	9.34551	8.93566	9.102723	(-0.095,-0.166,49.646)
11	0	0	0	0	(-0.095,-0.166,49.646)
12	0	0	0	0	(-0.095,-0.166,49.646)

- The better performance is achieved with the increased number of processes owing to distributed memory parallelism, here 10 processors gave the best result.
- The processors 3, 6, 7, 9, 11 and 12 processors failed to produce the timing results due to uneven work distribution of 20,000 particles, which meant to say that when it is not uniformly distributed among the processors, some of them can run it workload and stay idle. It could also be due to memory constraints as well.
- Also, for processor 7 onwards the execution command changed to mpirun – use-hwthread-cpus-np 7./MPI.exe.
  The hyperthread or hardware thread are launched to improve the performance.

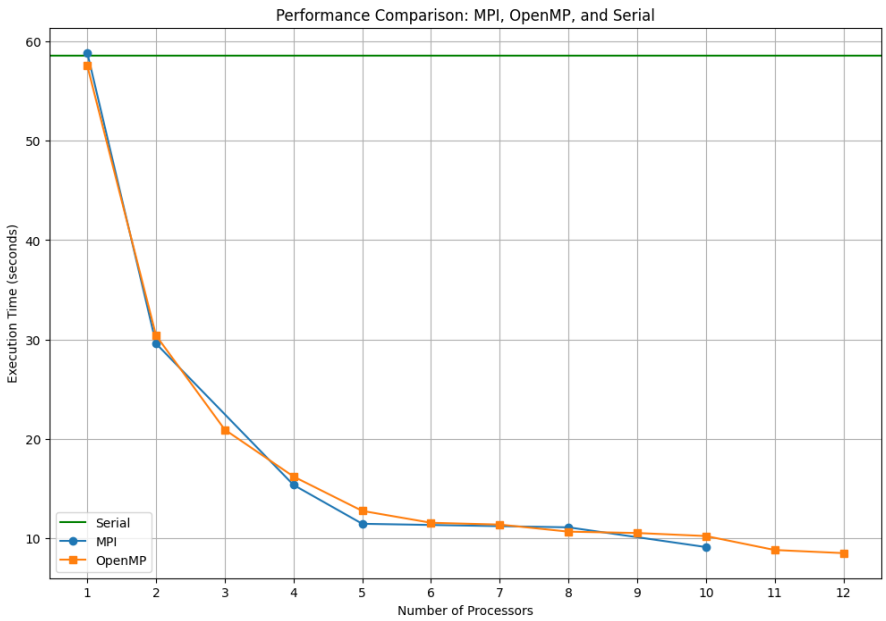
OpenMP is another parallel programming which much simpler compared to the previous MPI which also make use of memory parallelism. Here also, the molecules are simulated by the processors at optimization level of 2. The command to compile OpenMp is found to be $ gcc -O2 -fopenmp md-OpenMP.c -lm -o md_openmp.exe. And its execution command is export OMP_NUM_THREADS ./md_openmp.exe. The execution command unlike MPI is same for all the processors. And it worked on all the processors. The simulation is executed on each number of processors thrice and its average is taken.

- The methodology for calculating the run time of the simulation is the 'time' command.
- The centre of mass remained the same for all.

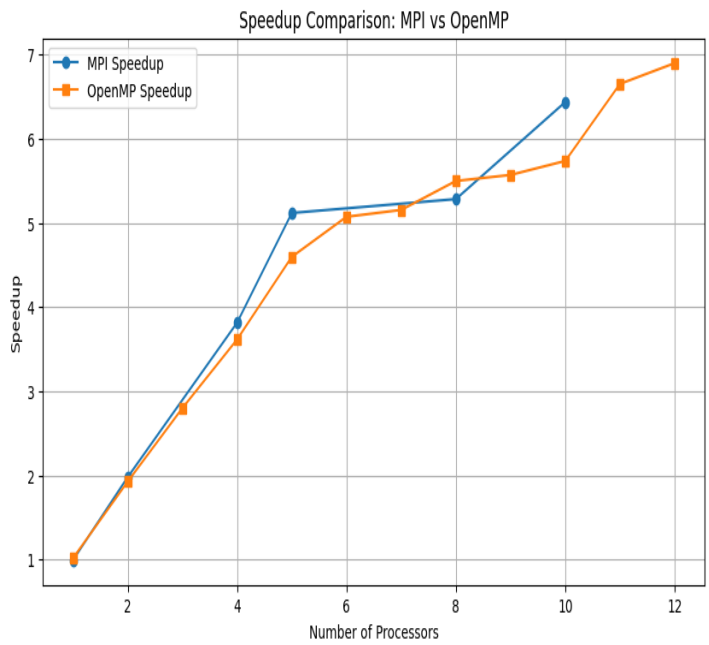
The timings along with Centre of mass for OpenMP is plotted below:

OPEN MP					
Processor	TimeStamp 1	TimeStamp 2	TimeStamp 3	AVG	Centre of Mass(x,y,z)
1	56.5524	59.6258	56.4957	57.55797	(-0.095,-0.166,49.646)
2	30.3987	30.662	30.141	30.40057	(-0.095,-0.166,49.646)
3	20.8829	20.7566	21.1241	20.9212	(-0.095,-0.166,49.646)
4	16.297	16.0951	16.2167	16.20293	(-0.095,-0.166,49.646)
5	11.4395	13.3884	13.4048	12.74423	(-0.095,-0.166,49.646)
6	11.5153	11.5076	11.6242	11.54903	(-0.095,-0.166,49.646)
7	11.5041	11.3438	11.2519	11.3666	(-0.095,-0.166,49.646)
8	10.4317	11.2389	10.2955	10.65537	(-0.095,-0.166,49.646)
9	10.4705	10.7164	10.3677	10.5182	(-0.095,-0.166,49.646)
10	10.4179	10.1333	10.0887	10.2133	(-0.095,-0.166,49.646)
11	9.13477	9.16761	8.12175	8.808043	(-0.095,-0.166,49.646)
12	8.37975	8.41318	8.68719	8.493373	(-0.095,-0.166,49.646)

- Here, the best performance was observed at 12 processors with avg time for execution of the task falling from 57.55 seconds for 1 processor to 8.49 seconds for 12 processors.



- The execution of serial programming acts as as base line at 58.6083 seconds.
- With the increase in processors, the execution time decreases.
- Just a jump to 2 processors, reduced the execution to half for MPI and OpenMP.
- Even with the introduction of 5 processors, MPI performed slightly better than OpenMP.
- Then the execution time of MPI values remained stagnant till the addition of 10 processors.
- However, with the addition of two more processors at 10, MPI's value decreased and was performing better than OpenMP.
- Eventually, when the max of 12 core is added, OpenMP turned to have better performance with a lowest execution time for the simulation.

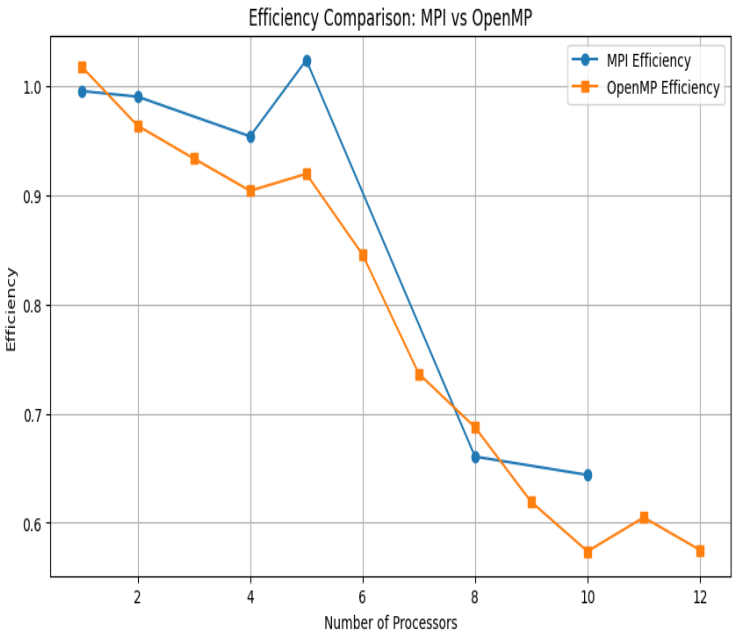


Speedup indicates how many times faster the parallel implementation is compared to the serial implementation.

For the first 5 processors, the speedup comparison of MPI is higher than OpenMP and after 5 to 8, the growth is very small, even on 10 processors, MPI is better, however, OpenMP is ended up at the best performing parallel program at 12 cores.

Efficiency is defined as the speedup divided by the number of processors. It gives an indication of how well the processors are being utilized in the parallel implementation.

Efficiency of MPI drops after 8 processors and Open MP ended up as the most efficient parallel computing system.



The findings of the report demonstrated the effectiveness of parallelization techniques in improving computational performance for the given particle simulation graph.

- The Serial Implementation acted as a baseline performance.
- MPI Implementation is best suitable for distributed systems but has constraints.
- OpenMP Implementation is found to be easy to implement and provides best performance among all and can work in all processors.

REFERENCES

1. Dongarra, J., Golub, G. H., Grosse, E., Moler, C. and Moore, K. (2008) 'Netlib and NA-Net: Building a Scientific Computing Community.' *IEEE Annals of the History of Computing*, 30(2) pp. 30–41.

APPENDIX

1. HPC data.xlsx
2. HPC.ipynb

Department of Computing and Mathematics

Big Data

Project Title:
"Comparative Analysis of Ride Durations from Baylis Road, Waterloo Station and Other Stations in 2014"

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STUDENT ID: 23645007

"I, Blessen Koshy Sujan, declare that the work titled “Comparative Analysis of Ride Durations from Baylis Road, Waterloo Station and Other Stations in 2014” is my own, except where acknowledged, and has not been submitted elsewhere."

NAME: BLESSEN KOSHY SUJAN
DATE: 16/05/2024

RESEARCH HYPOTHESIS: **“Rides in 2014 starting from Baylis Road, Waterloo station were shorter compared to other stations”**

INTRODUCTION

This report uses PySpark in a Google Colab to conduct a thorough analysis of cycle hire data from 2014. This dataset provided important insights into the trends and cycle hire durations from various stations in London. The objective is to reveal whether the hypothesis of Baylis road rides is shorter compared to other rides.

INITIALIZATION

Apache Spark is installed in Colab Notebook as the initial step. The process involved are downloading and extracting Spark binaries, installing Java dependencies, and configuring environment variables to facilitate smooth data processing and analysis.

LOADING AND CLEANING THE DATA

The Transport for London (TfL) open data repository is accessed to obtain the 2014 bicycle ride dataset. The dataset is downloaded, unzipped, and then loaded into a Spark Data Frame called "bike."

The dataset comprised 9 columns: "Rental ID, Duration, Bike ID, End Date, EndStation ID, Start Date, StartStation ID, and StartStation Name" and 11481596 rows of data. Data cleaning operations are executed to ensure data integrity and reliability. This includes the dropping of rows with missing values, dropping duplicates and conversion of the "Duration" column to integers, and filtering out of rides with durations exceeding 2.5 hours (9000 seconds) as 99.99 % of rides are under this hour. This is because I have made up a null hypothesis (H0), that the rides were under 2.5 hours and an alternative hypothesis (H1), that the rides are not.

A new data set named ‘bike cleaned’ is created. After filtering out, this size is reduced to 10124059 values. Therefore, the final data is made up of 10124059 data points. Then, the values with Zero seconds are removed as it would represent zero rides. Additionally, a histogram is generated to visualize the distribution of ride durations, thereby providing valuable insights into the variability and range of ride durations across the dataset.

summary	Duration
count	10242483
mean	1466.081447242822
stddev	12946.838147599005
min	-1020
25%	480.0
50%	840.0
75%	1320.0
max	99900

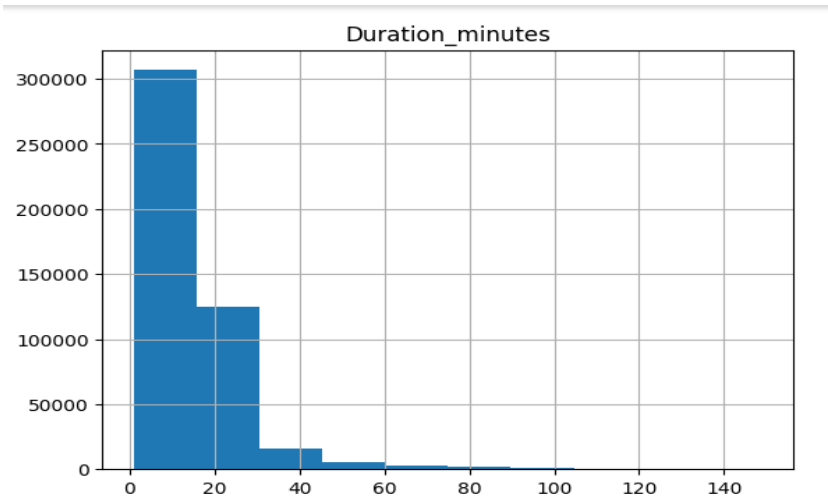
Before Cleaning

summary	Duration
count	10124059
mean	1073.9097628727766
stddev	1001.3147336849778
min	60
25%	480
50%	840
75%	1320
max	9000

After Cleaning

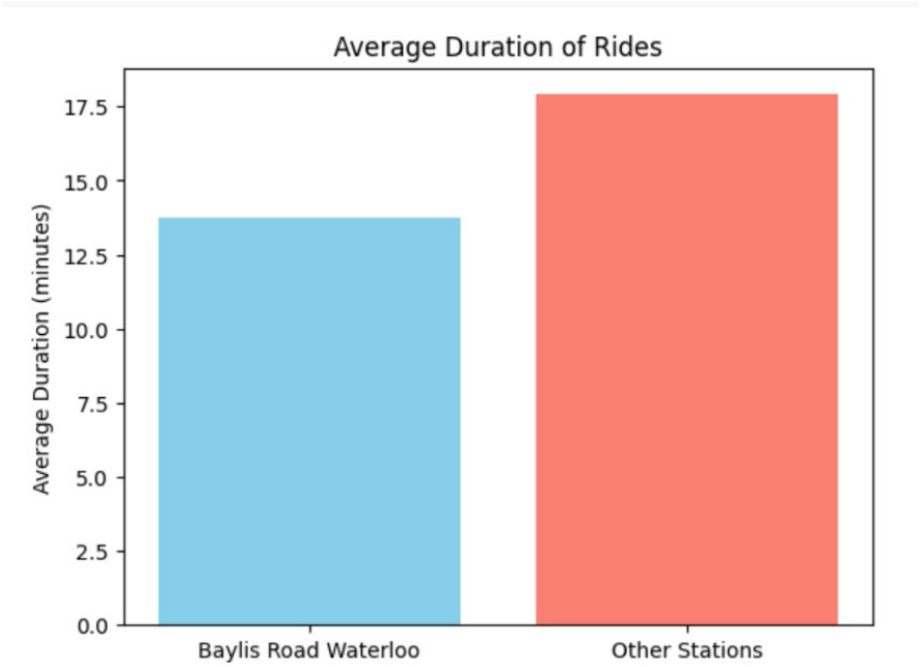
In parallel with our analysis, a meticulous quality check is performed on the dataset to ensure its reliability and consistency. This includes the verification of missing values in the duration column and an examination of data formats to confirm adherence to expected standards. Summary statistics for the duration column are utilized to provide additional insights into the distribution and characteristics of ride durations.

Having completed the cleaning of the dataset, the analysis phase of our journey is embarked upon. Summary statistics are computed for ride durations, encompassing essential metrics such as the minimum, mean, maximum, and skewness. This statistical overview offers a comprehensive understanding of the distribution and characteristics of ride durations within our dataset. Furthermore, ride durations are converted from seconds to minutes to facilitate a more intuitive representation of temporal aspects and the column duration in minutes is converted into integer type. It is found that the mean of the data set is 17.89 minutes and the minimum and maximum duration of the rides in minutes are 1 min and 150 min respectively.



ANALYSIS AND TESTING

Segmentation of the dataset based on ride origin is carried out, distinguishing rides originating from Baylis Road, Waterloo Station, from those originating from other stations. Average durations for rides from Baylis Road, Waterloo Station, are calculated and contrasted against the average durations for rides from other stations. The number of rides from Baylis road is found to be 27064 with an average duration of about **13.75** minutes. At the same times, the rest of the rides from other stations has an average ride duration of about **17.90** minutes. Through the visualization of this comparison via a bar plot, valuable insights are gained into potential disparities in ride durations across different stations, laying the groundwork for further exploration and analysis. This supported the given hypothesis that **Rides in 2014 starting from Baylis Road, Waterloo station were shorter compared to other stations**.



To ascertain the significance of observed differences in average ride durations, a rigorous statistical testing approach is utilized. A one-tailed t-test is conducted between rides starting from Baylis Road, Waterloo Station, and rides originating from other stations. It is implemented as there are only two groups and the test is capable if there is equal variance or not.

A one-tailed test is suitable when the hypothesis specifies a direction for the expected effect. Based on our theory, rides from Baylis Road Waterloo Station are generally shorter than those from other stations.
Null Hypothesis (H0): it states that the average ride duration from Baylis Road Waterloo Station is greater than or equal to the average ride duration from other stations.
Alternative Hypothesis (H1): It states that the average ride duration from Baylis Road Waterloo Station is less than the average ride duration from other stations.

The T-statistic is -55.581, which indicates that the sample mean of the ride durations from Baylis Road Waterloo Station is substantially lower than that from other stations. The p-value is 0.0 which is a near impossible value, however, it represents the value is very much less than the small threshold and it conveys the result is highly statistically significant.

We reject the null hypothesis because the one-tailed P-value is significantly less than 0.05. This gives strong proof that rides from Baylis Road Waterloo Station tend to be much shorter than those at other stops.

CONCLUSION
Thus, it is proved by both analysis and hypothesis test that the cycle rides from Baylis road, Waterloo Station is shorter than all other stations.

The Role of Big Data in Achieving a Sustainable Global Environment

INTRODUCTION

Big data analytics is widely recognized as a transformative force for solving environmental issues and promoting sustainability goals around the world. Many contributions to the environmental trends can be obtained, while the management of resources can be enhanced, and appropriate measures to decrease the amount of carbon dioxide and other dangerous elements released to the atmosphere may be determined by firms and administrations that analyze massive data amounts from disparate sources.

SOCIETAL

The inclusion of big data in sustainability efforts has the possibility for delivering significant social benefits. Both governments and businesses may adapt efforts to better meet the needs of communities through the analysis of data on energy consumption patterns, travel preferences, and environmental conditions (Harford, 2014). People can employ big data analytics to make better informed decisions about how their actions influence the environment. People may take part in sustainability activities in their daily life by using applications and platforms which provide them with customised feedback on their energy consumption, waste production, and carbon footprint (Harford, 2014). Big data technologies, by boosting awareness and advocating changes in behaviour, support ecological consciousness and grassroots campaigning.

LEGAL

From the perspective of law, the application of big data in conservation efforts and reduction of carbon projects increases vital legal issues. Governments need to balance the significance of data-driven interventions on privacy, data protection, and civil liberties concerns (European Parliament and Council, 2016; California Legislative Information, 2018). The legal framework regarding data ownership and access serves an essential part in deciding the efficiency of sustainability initiatives. Open data policies and data-sharing agreements facilitate collaboration between scientists, policymakers, and stakeholders in the field of environmental monitoring and climate research.

ETHICAL

When using big data technology for sustainability purposes, ethical considerations must come first. Respecting the rights and dignity of people, communities, and ecosystems is a necessity when collecting, analysing, and using environmental data (Floridi & Taddeo, 2016). Environmental inequalities and social disparities can be made more severe by biases found in datasets or algorithms, which might result in an uneven distribution of benefits and costs related to the degradation of the environment. When creating and carrying out massive records structures, businesses must prioritise equity, fairness, and diversity first to guarantee that sustainability efforts benefit every member of humanity.

CONCLUSION

In conclusion, a pivotal role is played by big data in achieving a more sustainable global environment through the provision of valuable insights, driving innovation, and enabling evidence-based decision-making (Manyika et al., 2011). However, the adoption of big data raises complex social, legal, and ethical considerations that must be addressed to maximize its potential for positive impact. By integrating principles of social responsibility, legal compliance, and ethical conduct into big data initiatives, the transformative power of data analytics can be harnessed by stakeholders to advance environmental sustainability and create a better future for generations to come.

REFERENCES

1. Manyika, J., Chui, M., Brown, B., Bughin, J., Dobbs, R., Roxburgh, C., & Byers, A. H. (2011). Big data: The next frontier for innovation, competition, and productivity. McKinsey Global Institute.
2. Boyd, D., & Crawford, K. (2012). Critical questions for big data: Provocations for a cultural, technological, and scholarly phenomenon. Information, Communication & Society, 15(5), 662-679.
3. Harford, T. (2014). Big data: A big mistake? Significance, 11(5), 14-19.
4. Floridi, L., & Taddeo, M. (2016). What is data ethics? Philosophical transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences, 374(2083), 20160360.
5. California Legislative Information. (2018). California Consumer Privacy Act of 2018. Retrieved from https://leginfo.legislature.ca.gov/faces/billTextClient.xhtml?bill_id=201720180AB375

APPENDIX

1. One drive link: BigData_Assessment (2).ipynb

2. Colab Link: https://colab.research.google.com/drive/1NfnfvMga-58Ku8TDQmJympaCeyYIkM0X?usp=sharing